

$$1. (a) \sum_{i=1}^{\infty} \frac{1}{i(i+2)(i+3)}$$

$$\text{Assume } \frac{1}{i(i+2)(i+3)} = \frac{A}{i} + \frac{B}{i+2} + \frac{C}{i+3}$$

$$\frac{A(i^2+6i+5)}{i(i+2)(i+3)} + \frac{B(i^2+3i)}{(i+2)(i+3)} + \frac{C(i^2+2i)}{i(i+2)} = \frac{1}{i(i+2)(i+3)}$$

$$\therefore (A+B+C)i^2 + (6A+3B+2C)i + 6A = 1$$

$$\therefore A = \frac{1}{6} \quad B = -\frac{1}{2} \quad C = \frac{1}{3}$$

$$\therefore \sum_{i=1}^{\infty} \frac{1}{i(i+2)(i+3)} = \sum_{i=1}^{\infty} \frac{1}{6i} - \sum_{i=1}^{\infty} \frac{1}{2(i+2)} + \sum_{i=1}^{\infty} \frac{1}{3(i+3)}$$

$$\text{Assume } H_N = \sum_{i=1}^N \frac{1}{i}, \text{ assume } j = i+2, k = i+3$$

$$\therefore \sum_{i=1}^N \frac{1}{i+2} = \sum_{j=3}^{N+2} \frac{1}{j} = H_{N+2} - 1 - \frac{1}{2} = H_N + \frac{1}{N+1} + \frac{1}{N+2} - \frac{3}{2}$$

$$\sum_{i=1}^{\infty} \frac{1}{i(i+2)(i+3)} \therefore \sum_{i=1}^N \frac{1}{i+3} = \sum_{k=4}^{N+3} \frac{1}{k} = H_{N+3} - 1 - \frac{1}{2} - \frac{1}{3} = H_N + \frac{1}{N+1} + \frac{1}{N+2} + \frac{1}{N+3} - \frac{11}{6}$$

$$= \lim_{N \rightarrow \infty} \sum_{i=1}^N \frac{1}{i(i+2)(i+3)} = \frac{1}{6} H_N - \frac{1}{2} (H_N + \frac{1}{N+1} + \frac{1}{N+2} - \frac{3}{2}) + \frac{1}{3} (H_N + \frac{1}{N+1} + \frac{1}{N+2} + \frac{1}{N+3} - \frac{11}{6})$$

$$= \frac{1}{6} H_N - \frac{1}{2} H_N + \frac{1}{2} H_N - \frac{1}{2} (\frac{1}{N+1} + \frac{1}{N+2}) + \frac{3}{4} + \frac{1}{3} (\frac{1}{N+1} + \frac{1}{N+2} + \frac{1}{N+3}) - \frac{11}{18}$$

when $N \rightarrow \infty$,

$$\sum_{i=1}^{\infty} \frac{1}{i(i+2)(i+3)} = \frac{3}{4} - \frac{11}{18} = \boxed{\frac{5}{36}}$$

$$1. (b) \sum_{i=1}^{\infty} (i+1)(0.1)^i \quad \text{Solve by differentiation.}$$

$$\text{we have } \sum_{i=1}^{\infty} x^i = \lim_{n \rightarrow \infty} \sum_{i=1}^n x^i = \lim_{n \rightarrow \infty} \frac{1-x^{n+1}}{1-x}$$

$$\text{if } x < 1, \sum_{i=1}^{\infty} x^i = \frac{1}{1-x}$$

$$\frac{d}{dx} \sum_{i=1}^{\infty} x^i = \frac{d}{dx} \frac{1}{1-x} \quad \text{LHS} = 1 \cdot x^0 + 2 \cdot x^1$$

$$\therefore \sum_{i=1}^{\infty} i x^{i-1} = \frac{1}{(1-x)^2} \quad \sum_{i=1}^{\infty} i x^i = 1 \cdot x^1 + 2 \cdot x^2 \dots = \frac{x}{(1-x)^2} \quad (1)$$

$$\sum_{i=1}^{\infty} (0.1)^i = \lim_{n \rightarrow \infty} \frac{a_1 - a_n q^n}{1 - a_1} = \frac{0.1}{0.9} = \frac{1}{9} \quad (2)$$

according to (1).

$$\sum_{i=1}^{\infty} i(0.1)^i = \frac{0.1}{(1-0.1)^2} = \frac{0.1}{0.81} = \frac{1}{8.1}$$

$$= \frac{1}{9} \times \frac{10}{9} = \frac{10}{81} \quad (3)$$

$$\therefore \sum_{i=1}^{\infty} (i+1)(0.1)^i = \sum_{i=1}^{\infty} i(0.1)^i + \sum_{i=1}^{\infty} (0.1)^i = (3) + (2)$$

$$= \frac{10}{81} + \frac{1}{9} = \boxed{\frac{19}{81}}$$

Question 2

Solve by Cholesky decomposition

Step ①: check if $A^T = A$? obviously.

$\therefore$ Cholesky decomposition accessible. Assume $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$ then $L^T = \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$

$$LL^T = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{31} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix} = \begin{bmatrix} 5 & 1 & -5 \\ 1 & 1 & -2 \\ -5 & -2 & 13 \end{bmatrix} \quad \text{And } l_{11}, l_{22}, l_{33} > 0$$

First Row: $l_{11}^2 = 5$, $l_{21}l_{11} = 1$, $l_{31}l_{11} = -5$

$\therefore l_{11} = \sqrt{5}$, $l_{21} = \frac{1}{\sqrt{5}}$, $l_{31} = -\sqrt{5}$

$$\therefore L = \begin{bmatrix} \sqrt{5} & 0 & 0 \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ -\sqrt{5} & -\frac{\sqrt{5}}{2} & \frac{3}{2}\sqrt{5} \end{bmatrix} \quad L^T = \begin{bmatrix} \sqrt{5} & \frac{1}{\sqrt{5}} & -\sqrt{5} \\ 0 & \frac{2}{\sqrt{5}} & -\frac{\sqrt{5}}{2} \\ 0 & 0 & \frac{3}{2}\sqrt{5} \end{bmatrix}$$

Second Row: $l_{21}^2 + l_{22}^2 = 1$, $l_{21}l_{31} + l_{22}l_{32} = -2$

$l_{22} = \frac{2}{\sqrt{5}}$, $l_{32} = -\frac{\sqrt{5}}{2}$

Third Row: $l_{31}^2 + l_{32}^2 + l_{33}^2 = 13$, $l_{33} = \frac{3}{2}\sqrt{5}$

Step ② Calculation. $Ax = b$ $LL^Tx = b$

Assume $y = L^Tx$, $Ly = b$, now the target to solve $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} \sqrt{5} & 0 & 0 \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ -\sqrt{5} & -\frac{\sqrt{5}}{2} & \frac{3}{2}\sqrt{5} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \begin{cases} \sqrt{5}y_1 = 1 \\ \frac{1}{\sqrt{5}}y_1 + \frac{2}{\sqrt{5}}y_2 = 1 \\ -\sqrt{5}y_1 - \frac{\sqrt{5}}{2}y_2 + \frac{3}{2}\sqrt{5}y_3 = 0 \end{cases} \quad \therefore \begin{cases} y_1 = \frac{1}{\sqrt{5}} \\ y_2 = \frac{2}{\sqrt{5}} \\ y_3 = \frac{4}{9\sqrt{5}} \end{cases}$$

Step ③ Find x by $L^Tx = y$, assume $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$\begin{bmatrix} \sqrt{5} & \frac{1}{\sqrt{5}} & -\sqrt{5} \\ 0 & \frac{2}{\sqrt{5}} & -\frac{\sqrt{5}}{2} \\ 0 & 0 & \frac{3}{2}\sqrt{5} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ \frac{4}{9\sqrt{5}} \end{bmatrix}$$

$$\begin{cases} \frac{3}{2}\sqrt{5}x_3 = \frac{4}{9\sqrt{5}} \\ \frac{2\sqrt{5}}{5}x_2 - \frac{\sqrt{5}}{2}x_3 = \frac{2}{\sqrt{5}} \\ \sqrt{5}x_1 + \frac{\sqrt{5}}{5}x_2 - \sqrt{5}x_3 = \frac{\sqrt{5}}{5} \end{cases} \quad \therefore \begin{cases} x_3 = \frac{8}{27} \\ x_2 = \frac{37}{27} \\ x_1 = \frac{2}{9} \end{cases}$$

$$\therefore x = \begin{bmatrix} \frac{2}{9} \\ \frac{37}{27} \\ \frac{8}{27} \end{bmatrix}$$

Question 3

(a) $A = L + U + D$

① G.S method, generally,

$(L+D)x^{k+1} = -Ux^k + b$

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 5 & 0 \\ 1 & 0 & 25 \end{bmatrix} \begin{bmatrix} x_1^{k+1} \\ x_2^{k+1} \\ x_3^{k+1} \end{bmatrix} = \begin{bmatrix} 0 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1^k \\ x_2^k \\ x_3^k \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad \therefore \text{update formula: } \begin{bmatrix} 2x_1^{k+1} \\ -3x_1^{k+1} + 5x_2^{k+1} \\ x_1^{k+1} + 25x_3^{k+1} \end{bmatrix} = \begin{bmatrix} 1 + 3x_3^k - x_3^k \\ 2 \\ 0 \end{bmatrix}$$

Iter. 1 $x^0 = [0, 0, 0]$

Iter. 2 $x^1 = [\frac{1}{2}, \frac{7}{10}, -\frac{1}{50}]$

$\therefore$ after 2 iter.,

$$\begin{cases} 2x_1^1 = 1 \\ -3x_1^1 + 5x_2^1 = 2 \\ 25x_3^1 + x_1^1 = 0 \end{cases} \quad \therefore \begin{cases} x_1^1 = \frac{1}{2} \\ x_2^1 = \frac{7}{10} \\ x_3^1 = -\frac{1}{50} \end{cases}$$

$$\begin{cases} 2x_1^2 = \frac{156}{50} \\ -3x_1^2 + 5x_2^2 = 2 \\ x_1^2 + 25x_3^2 = 0 \end{cases} \quad \therefore \begin{cases} x_1^2 = \frac{39}{25} = 1.56 \\ x_2^2 = \frac{167}{125} = 1.336 \\ x_3^2 = -\frac{39}{625} = -0.0624 \end{cases}$$

$x = [1.56, 1.336, -0.0624]$

② Jacobi method $Dx^{k+1} = -Ux^k - Lx^k + b$

updating formula:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 25 \end{bmatrix} \begin{bmatrix} x_1^{k+1} \\ x_2^{k+1} \\ x_3^{k+1} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1^k \\ x_2^k \\ x_3^k \end{bmatrix} - \begin{bmatrix} -3 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1^k \\ x_2^k \\ x_3^k \end{bmatrix}$$

$$\begin{cases} 2x_1^{k+1} = 1 + 3x_2^k - x_3^k \\ 5x_2^{k+1} = 2 + 3x_1^k \\ 25x_3^{k+1} = -x_1^k \end{cases}$$

Iter. 1

Iter. 2

$$\begin{cases} 2x_1^1 = 1 \\ 5x_2^1 = 2 \\ 25x_3^1 = 0 \end{cases} \quad \begin{cases} x_1^1 = \frac{1}{2} \\ x_2^1 = \frac{2}{5} \\ x_3^1 = 0 \end{cases} \quad \therefore x^1 = [\frac{1}{2}, \frac{2}{5}, 0]$$

$$\begin{cases} 2x_1^2 = \frac{11}{5} \\ 5x_2^2 = \frac{7}{2} \\ 25x_3^2 = -\frac{1}{2} \end{cases} \quad \begin{cases} x_1^2 = \frac{11}{10} \\ x_2^2 = \frac{7}{10} \\ x_3^2 = -\frac{1}{50} \end{cases} \quad \therefore x^2 = [\frac{11}{10}, \frac{7}{10}, -\frac{1}{50}]$$

(b) G.S method iterative formula $x^{k+1} = Hx^k + b$, $H = -(D+U)^{-1}U = -\begin{bmatrix} 2 & 0 & 0 \\ -3 & 5 & 0 \\ 1 & 0 & 25 \end{bmatrix}^{-1} \begin{bmatrix} 0 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 5 & 0 \\ 1 & 0 & 25 \end{bmatrix} \begin{bmatrix} x_{11} & 0 & 0 \\ x_{21} & x_{22} & 0 \\ x_{31} & x_{32} & x_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases} 2x_{11} = 1 \\ -3x_{11} + 5x_{21} = 0 \\ x_{11} + 25x_{31} = 0 \end{cases} \quad \begin{cases} 5x_{22} = 1 \\ 25x_{32} = 0 \\ x_{22} = \frac{1}{5} \\ x_{32} = 0 \end{cases} \quad \begin{cases} 25x_{33} = 1 \\ x_{33} = \frac{1}{25} \end{cases}$$

$$\therefore H = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ \frac{3}{10} & \frac{1}{5} & 0 \\ -\frac{1}{50} & 0 & \frac{1}{25} \end{bmatrix} \begin{bmatrix} 0 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & \frac{3}{10} & -\frac{3}{10} \\ 0 & -\frac{3}{50} & \frac{1}{50} \end{bmatrix}$$

$$H - I\lambda = \begin{bmatrix} -\lambda & \frac{3}{2} & -\frac{1}{2} \\ 0 & \frac{3}{10} - \lambda & -\frac{3}{10} \\ 0 & -\frac{3}{50} & \frac{1}{50} - \lambda \end{bmatrix}$$

Assume eigenvalues of H are λ . $\det(H - I\lambda) = 0$

$$\Rightarrow \frac{1}{50} \left(\frac{3}{10} - \lambda \right) + \left(\frac{1}{50} - \lambda \right) \left(\lambda - \frac{3}{10} \right) = 0$$

We solve $\lambda_1 = 0$, $\lambda_2 = 0$, $\lambda_3 = 0.92$

$\rho(H) < 1$. $\therefore$ G.S method can converge from any starting point.

Question 3 (cont.)

(c). Jacobi method iteration formula: $x^{k+1} = Hx^k + b$ $H = -D^{-1}(L+u)$

$$D^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & \frac{1}{25} \end{bmatrix} \quad L+u = \begin{bmatrix} 0 & -3 & 1 \\ -3 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & \frac{1}{25} \end{bmatrix} \begin{bmatrix} 0 & 3 & -1 \\ 3 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ \frac{3}{5} & 0 & 0 \\ -\frac{1}{25} & 0 & 0 \end{bmatrix}$$

Assume eigenvalues of H are λ .

$$\det(H - \lambda I) = \begin{vmatrix} -\lambda & \frac{3}{2} & -\frac{1}{2} \\ \frac{3}{5} & -\lambda & 0 \\ -\frac{1}{25} & 0 & -\lambda \end{vmatrix} = -\frac{1}{25}(-\frac{1}{2}\lambda) - \lambda(\lambda^2 - \frac{9}{10}) = 0$$

$$= \frac{1}{50}\lambda + \lambda(\frac{9}{10} - \lambda^2) = 0$$

$$\lambda_1 = 0, \quad \lambda_2 = \sqrt{0.92} \quad \lambda_3 = -\sqrt{0.92}$$

$$\approx 0.96 \quad \approx -0.96$$

$$\rho(H) = 0.96 < 1$$

$\therefore$ Jacobi method can converge from $[0, 0, 0]^T$

Question 4

The Statement is false.

For example:

$$A = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 0 & 0 \\ 0 & 9 \end{bmatrix}$$

Assume the eigenvalue of $A^T A$ is λ .

$$\lambda_1 = 0,$$

$$\therefore \det \begin{pmatrix} -\lambda & 0 \\ 0 & 9-\lambda \end{pmatrix} = -9\lambda + \lambda^2 = 0, \quad \lambda_2 = 9$$

$$\therefore \|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sqrt{9} = 3 > 2.$$

If we take $c = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, The system:

$$x^{k+1} = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} x^k + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From a point $x^0 = \begin{bmatrix} a \\ b \end{bmatrix}$

$$x^1 = \begin{bmatrix} 3b \\ 0 \end{bmatrix}, \quad x^2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ convergent!}$$

$\therefore$ for $x = Hx + c$, when $H = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$, and $c = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, satisfies $\|H\|_2 > 2$, but will eventually converge.

Question 5

Assume $f(x) = x^4 - 7$, then $7^{\frac{1}{4}}$ is the root of $f(x) = 0$.

First, we find the range of x^* . $f(x)$ is obviously monotone increasing when $x > 0$

$f(1.6) = (1.6)^4 - 7 = -0.446 < 0$, $f(1.7) = (1.7)^4 - 7 = 1.352 > 0$
 $\therefore x^* \in [1.6, 1.7]$

Then, we define I . assume $d = 1.7 - x^*$
 $I = [2x^* - 1.1, 1.1] \subseteq [1.5, 1.7]$

According to Newton's Method. $g(x) = x - \frac{f(x)}{f'(x)} = x - \frac{x^4 - 7}{4x^3} = \frac{4}{7}x^3 + \frac{7}{4}x$
 $g'(x) = \frac{3}{4} - \frac{21}{4}x^{-4}$. (Also monotone increasing, then extreme value exists in edge.)
 $|g'(1.5)| = |\frac{3}{4} - \frac{21}{4} \times (1.5)^{-4}| \approx 0.2874 < 1$
 $|g'(1.7)| = |\frac{3}{4} - \frac{21}{4} \times (1.7)^{-4}| \approx 0.1214 < 1$

$\therefore$ Take any value in I , NM can converge to a certain value.

$x_1 = x_0 - \frac{x_0^4 - 7}{4x_0^3} = 1.7 - \frac{(1.7)^4 - 7}{4 \times (1.7)^3} \approx 1.63120$
 $x_2 = 1.63120 - \frac{(1.63120)^4 - 7}{4 \times (1.63120)^3} \approx 1.62660$
 $x_3 = 1.62660 - \frac{(1.62660)^4 - 7}{4 \times (1.62660)^3} \approx 1.62658$

$7^{\frac{1}{4}}$ converges to 1.62658.

Question 6

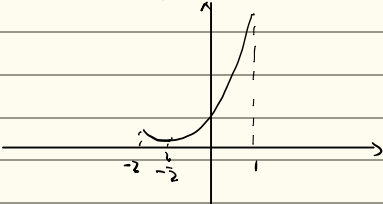
(a) $x_0 = -2, f_0 = 5e^{-4} \quad (-2, 5e^{-4})$
 $x_1 = 0, f_1 = 1 \quad (0, 1)$
 $x_2 = 1, f_2 = 2e^2 \quad (1, 2e^2)$
 $\therefore p_2(x) = f_0 l_0(x) + f_1 l_1(x) + f_2 l_2(x)$
 $l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{x(x-1)}{6} = \frac{1}{6}x(x-1) = \frac{1}{6}x^2 - \frac{1}{6}x$
 $l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x+2)(x-1)}{2 \cdot (-1)} = -\frac{1}{2}(x+2)(x-1) = -\frac{1}{2}x^2 - \frac{1}{2}x + 1$
 $l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x+2)x}{3 \cdot 1} = \frac{1}{3}(x+2)x = \frac{1}{3}x^2 + \frac{2}{3}x$

$p_2(x) = \frac{5}{6}e^{-4}(x^2 - x) + (1 - \frac{1}{2}x^2 - \frac{1}{2}x) + \frac{2e^2}{3}(x^2 + 2x)$

for $2x^2 + 6x + 5 = 0$,

(b) $\Delta = 36 - 4 \times 2 \times 5 < 0$, no solution.

$g(x) = 2x^2 + 6x + 5$ graph: $\max |2x^2 + 6x + 5| = |g(x)|$
 $= 3$



(c) $|f(x) - p_2(x)|$
 $= \frac{1}{3!} f^{(3)}(\xi) (x+2) \times (x-1)$
 $\xi \in [-2, 1]$
 $f(x) = (x^2 + 1)e^{2x}$
 $f'(x) = 2xe^{2x} + 2(x^2 + 1)e^{2x} = (2x^2 + 2x + 2)e^{2x}$
 $f''(x) = 2(2x^2 + 2x + 2)e^{2x} + (4x + 2)e^{2x} = (4x^2 + 8x + 6)e^{2x}$
 $f^{(3)}(x) = 2(4x^2 + 8x + 6)e^{2x} + (8x + 8)e^{2x} = (8x^2 + 24x + 20)e^{2x}$
 $|f(x) - p_2(x)| \leq \frac{1}{6} \max_{-2 \leq x \leq 1} |8x^2 + 24x + 20| e^{2x} \cdot \max_{-2 \leq x \leq 1} |(x+2)(x-1)|$
For $\oplus$: when $x = 1$ term $\oplus = 52e^2 \approx 384.231$
For $\ominus$: $\max_{-2 \leq x \leq 1} |8x^2 + 24x + 20| = 13$ at $x = -1$.
 $\therefore |f(x) - p_2(x)| \leq \frac{1}{6} \times 384.231 \times 2.112 \approx 135.27 < 136!$

